RAG Learning

OCR B A-Level Chemistry — Practical Activities

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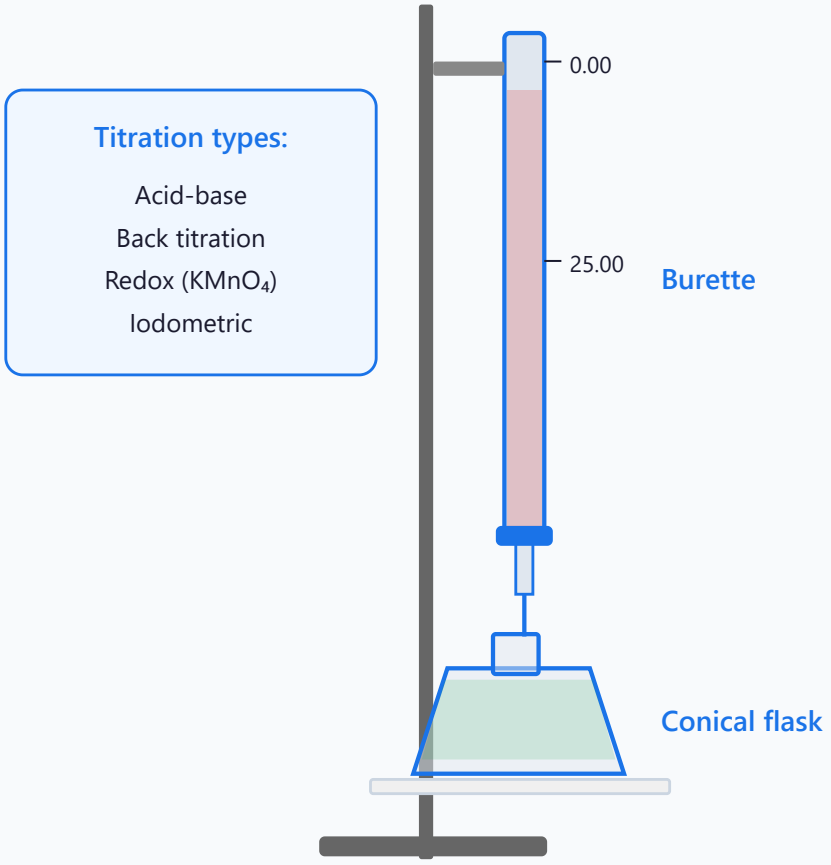
⚗ 1: Volumetric Analysis / Titration



Apparatus

- 50 cm^3 burette, clamp and stand
- 25.0 cm^3 pipette and filler
- Conical flasks (250 cm^3), white tile
- Suitable indicator or self-indicating reagent
- Volumetric flask (for preparing standard solutions)

Method

1. **Acid-base:** Pipette 25.0 cm^3 of one solution into a conical flask. Add indicator. Titrate from the burette until permanent colour change. Repeat for concordant results (within 0.10 cm^3).
2. **Back titration:** React the sample with known excess of reagent. Titrate the remaining unreacted reagent. Calculate original amount by difference. Useful when the analyte is insoluble (e.g., $CaCO_3$ in antacid tablets).

3. **Redox ($KMnO_4$):** Pipette acidified Fe^{2+} into flask. Titrate with $KMnO_4$ until permanent faint pink (self-indicating). Use H_2SO_4 to acidify (not HCl or HNO_3).
4. Calculate using n = cV and stoichiometric ratios from balanced equations.

Safety

- Acids and alkalis — corrosive/irritant, goggles.
- $KMnO_4$ — oxidising agent, stains; wear gloves.
- Use pipette filler — never mouth-pipette.

Analysis

Calculate: Mean concordant titre. n(known) = c × V. Use stoichiometry for n(unknown). c = n/V.

Back titration: n(original) = n(added) - n(remaining after back-titration).

Error Analysis

Source of Error	Type	Improvement
Overshooting end point	Systematic	Add drop-wise near end point
Parallax reading burette	Random	Eye level; bottom of meniscus (top for $KMnO_4$)
Not rinsing burette correctly	Systematic	Rinse with solution to be used
Including rough titre in mean	Systematic	Only average concordant results; exclude rough

Exam Tip: For back titrations, always set up the calculation as: moles reacted with analyte = total moles added − moles determined by titration. Common example: finding % $CaCO_3$ in limestone — add excess HCl, then titrate remaining HCl with NaOH.

Worked Example — Back Titration:

1.50 g of impure $CaCO_3$ was dissolved in 50.0 cm^3 of 1.00 mol dm^{-3} HCl (excess). The remaining HCl required 24.8 cm^3 of 0.500 mol dm^{-3} NaOH to neutralise it.

n(HCl total) = 1.00 × 0.0500 = 0.0500 mol

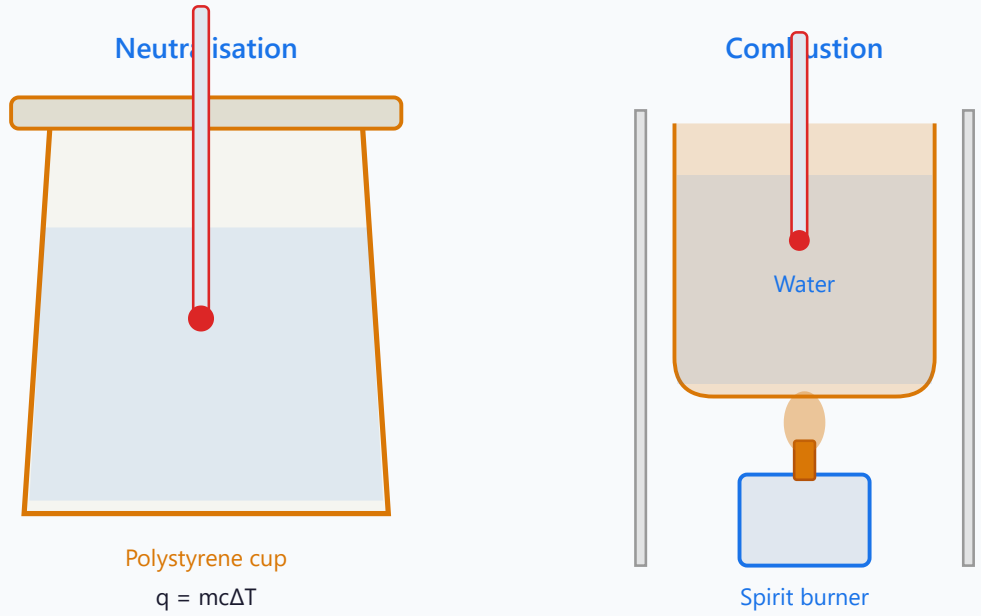
n(NaOH) = 0.500 × 0.0248 = 0.0124 mol = n(HCl remaining)

n(HCl reacted with $CaCO_3$) = 0.0500 − 0.0124 = 0.0376 mol

n($CaCO_3$) = 0.0376 / 2 = 0.0188 mol (from $CaCO_3$ + 2HCl equation)

Mass $CaCO_3$ = 0.0188 × 100.1 = 1.88 g... but we only had 1.50 g. Check: the $CaCO_3$ is impure, so the actual pure $CaCO_3$ mass = 0.0188 × 100.1 = 1.88 g — this exceeds the sample mass, indicating an error. Recalculating with correct titre would give a realistic % purity.

⚗ 2: Enthalpy Determination by Calorimetry



Neutralisation: ΔH ≈ -57 kJ mol^{-1} (strong acid + strong base)

Combustion: ΔH = -q/n (always less exothermic than literature due to heat loss)

Apparatus

- **Neutralisation:** Polystyrene cup, lid, thermometer, measuring cylinder, stopwatch

- **Combustion:** Spirit burner, copper calorimeter, thermometer, balance, draught shield, measuring cylinder

Method (Neutralisation)

1. Measure 25 cm^3 of acid into the polystyrene cup. Record temperature for 3 minutes.
2. At minute 4, add 25 cm^3 of alkali. Stir. Record temperature for 6+ minutes.
3. Use graphical extrapolation to find ΔT. Calculate $q = mc\Delta T$. Then $\Delta H = -q/n$.

Method (Combustion)

1. Measure 100 cm^3 of water into the calorimeter. Record initial temperature.
2. Weigh the spirit burner. Light it and heat the water until temperature rises ~20 °C.
3. Extinguish, reweigh. Calculate mass burned, $q = mc\Delta T$, $n = mass/M_r$. $\Delta H = -q/n$.

Safety

- Acids/alkalis — corrosive, goggles.
- Alcohols are flammable — keep away from other ignition sources.
- Hot apparatus — use tongs.

Analysis

Plot: Temperature vs. time for extrapolation (neutralisation).

Compare: Experimental vs. literature ΔH values. Discuss reasons for discrepancy.

Error Analysis

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic	Lid, insulation, draught shield, extrapolation
Incomplete combustion	Systematic	Ensure good air supply; clean wick
Evaporation of alcohol	Systematic	Cap promptly; weigh quickly
Heat capacity assumptions	Systematic	Acknowledged; dilute solution ≈ water

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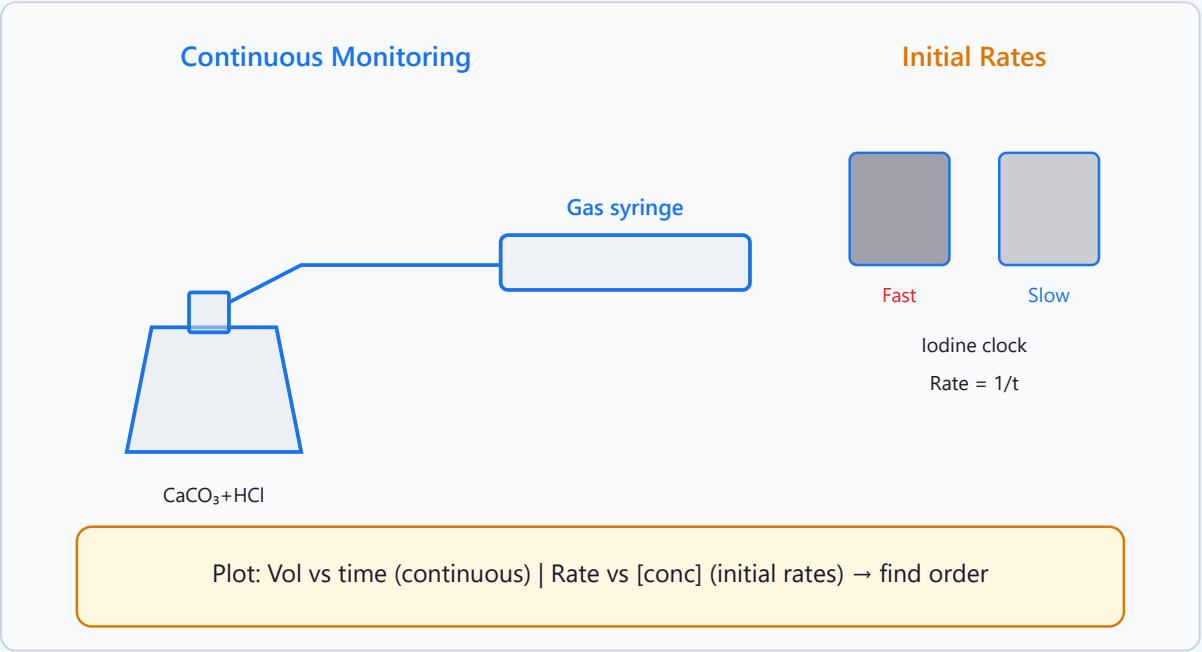
Exam Tip: The experimental ΔH(combustion) is always less negative (less exothermic) than the data book value. The main reason is heat loss to surroundings. Other factors include incomplete combustion and heat absorbed by the apparatus. Always state these clearly for full marks.

⚗ 3: Rates of Reaction



Apparatus

- **Continuous:** Conical flask, gas syringe, delivery tube, stopwatch, measuring cylinder
- **Initial rates:** Beakers/flasks, stopwatch, measuring cylinders, iodine clock reagents + starch

Method

1. **Continuous monitoring:** Collect gas at timed intervals from a reaction (e.g., $CaCO_3$ + HCl). Plot volume vs. time. Initial rate = gradient at t = 0.
2. **Initial rates (iodine clock):** Vary [one reactant], keep total volume constant. Time the colour change. Rate = 1/t. Plot rate vs. [varied species] to determine order.

3. Repeat with temperature variation to investigate Arrhenius relationship.

Safety

- HCl is irritant — goggles.
- SO_2 from thiosulfate reactions — ventilation.

Analysis

Continuous: Tangent at t = 0 gives initial rate. Compare different conditions.

Initial rates: Rate vs. [X]: straight line through origin = 1st order; parabola = 2nd order.

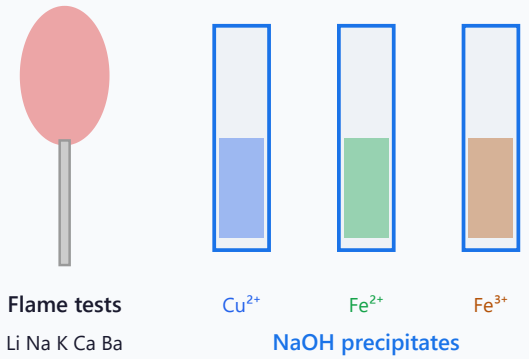
Arrhenius: ln(rate) vs. 1/T gives straight line; gradient = $-E_a/R$.

Error Analysis

Source of Error	Type	Improvement
Gas lost before sealing	Systematic	Seal quickly
Subjective colour change	Random	Same observer; colorimeter
Temperature variation	Systematic	Thermostatted water bath
Volume measurement error	Random	Use burettes for better precision

Exam Tip: To determine order from initial rate data: if doubling [X] doubles rate, the reaction is first order in X. If doubling [X] quadruples rate, it is second order. Zero order means the rate is unaffected by changing [X]. The overall order is the sum of individual orders.

⚗ 4: Qualitative Analysis



Anion Tests

Cl^-: $AgNO_3$ → white ppt

Br^-: $AgNO_3$ → cream ppt

I^-: $AgNO_3$ → yellow ppt

SO_4^{2-}: $BaCl_2$ → white ppt

CO_3^{2-}: HCl → CO_2 (limewater)

NH_4^+: NaOH + heat → NH_3

Always acidify with HNO_3 before $AgNO_3$ and $BaCl_2$ tests

Apparatus

- Test tubes, nichrome wire, Bunsen burner
- NaOH(aq), NH_3(aq), $AgNO_3$(aq), $BaCl_2$(aq), dilute HNO_3, HCl
- Limewater, litmus paper

Method

1. **Flame tests:** Clean wire → dip in sample → observe colour in Bunsen flame.
2. **Cation tests:** Add NaOH — observe precipitate colour and behaviour in excess.
3. **Halide tests:** Acidify with HNO_3, add $AgNO_3$. Confirm with NH_3: AgCl dissolves in dilute NH_3, AgBr in concentrated NH_3, AgI does not dissolve.
4. **Sulfate:** Acidify with HNO_3, add $BaCl_2$ → white precipitate.
5. **Carbonate:** Add HCl → effervescence, gas turns limewater milky.
6. **Ammonium:** Add NaOH, warm → NH_3 turns damp red litmus blue.

Safety

- $AgNO_3$ stains — gloves. $BaCl_2$ toxic — wash hands.
- Conc. HCl and NH_3 — fume cupboard.

Analysis

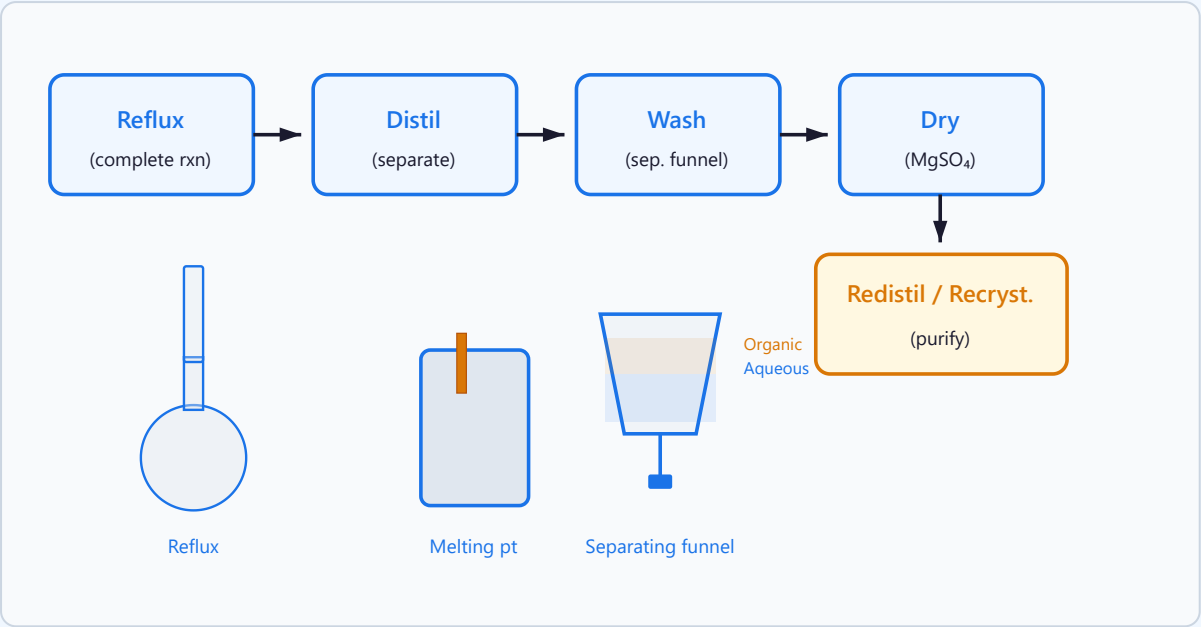
Record: Observations table for each test. Match results to identify unknown ions.

Error Analysis

Source of Error	Type	Improvement
Contamination	Systematic	Clean all equipment; use separate droppers
Na^+ masking flame colours	Systematic	View through cobalt blue glass
Not acidifying before tests	Systematic	Always acidify with HNO_3
Colour confusion (cream vs white)	Random	Compare alongside known standards

Exam Tip: To distinguish between halide silver precipitates further: add dilute NH_3 — AgCl dissolves (soluble complex forms); add conc. NH_3 — AgBr dissolves. AgI does not dissolve in either. This is a common extension question.

⚗ 5: Organic Preparation



Apparatus

- Round-bottomed flask, anti-bumping granules
- Reflux condenser (vertical) and distillation condenser (angled)
- Separating funnel, conical flask
- Drying agent (anhydrous $MgSO_4$ or Na_2SO_4)
- Buchner funnel + vacuum pump (for solids)
- Melting point apparatus, thermometer

Method

1. **Reaction:** Heat reaction mixture under reflux for the required time.
2. **Separation (liquids):** Distil to collect crude product. Wash in separating funnel with Na_2CO_3 (removes acid) then water. Dry with anhydrous drying agent. Filter. Redistil at correct boiling point.
3. **Purification (solids):** Dissolve crude solid in minimum hot solvent. Hot filter. Cool to crystallise. Buchner filter. Wash with cold solvent. Dry.
4. **Test purity:** Melting point (solids — sharp mp = pure) or boiling point (liquids).
5. Calculate percentage yield.

Safety

- Organic chemicals are flammable — electric heating, no flames.
- Work in fume cupboard.
- Vent separating funnel to release pressure.

Analysis

Calculate: % yield = (actual/theoretical) × 100. % atom economy = (M_r desired product / total M_r all products) × 100.

Error Analysis

Source of Error	Type	Improvement
Product lost in transfers	Systematic	Minimise transfers; rinse with small amounts
Incomplete reaction	Systematic	Reflux longer; use excess reagent
Product still wet	Systematic	Dry thoroughly; use desiccator
Wrong fraction collected	Systematic	Monitor thermometer; collect narrow range

Exam Tip: Always explain the purpose of each step: reflux = complete reaction; distillation = separate by boiling point; washing with Na_2CO_3 = removes acidic impurities; drying agent = removes water; recrystallisation = removes soluble impurities from solid. Each step reduces yield, so high yields require careful technique.

⚗ 6: Functional Group Identification Tests

Alkene (C=C)
Br_2(aq) decolourises
Orange → colourless

Aldehyde (CHO)
Tollens' → Ag mirror
Fehling's → red ppt

Carbonyl (C=O)
2,4-DNP → orange ppt
(aldehydes + ketones)

Alcohol (OH)
$K_2Cr_2O_7/H^+$ warm
Orange → green (1°/2°)

Carboxylic acid
Na_2CO_3 → fizzes (CO_2)
Limewater goes milky

Methyl ketone
Iodoform (I_2/NaOH)
Yellow ppt of CHI_3

Warm Tollens'/Fehling's/$K_2Cr_2O_7$ in a water bath at 60-70 °C

Apparatus

- Test tubes, water bath (60-70 °C)
- Bromine water, Tollens' reagent, Fehling's A+B, 2,4-DNP solution
- Acidified $K_2Cr_2O_7$, Na_2CO_3 solution, I_2/NaOH

Method

1. **C=C:** Add Br_2(aq) — decolourises = alkene.
2. **CHO (aldehyde):** Tollens' (silver mirror) or Fehling's (brick-red precipitate).
3. **C=O (carbonyl):** 2,4-DNP — orange precipitate = aldehyde or ketone.
4. **OH (alcohol 1°/2°):** Warm with acidified $K_2Cr_2O_7$ — orange to green.
5. **COOH:** Add Na_2CO_3 — effervescence = carboxylic acid.
6. **CH_3CO- (methyl ketone):** Iodoform test — yellow precipitate.

Safety

- $K_2Cr_2O_7$ is carcinogenic — gloves, fume cupboard.
- Tollens' — prepare fresh, dispose immediately (explosive on standing).
- Bromine water is toxic and corrosive — fume cupboard.

Analysis

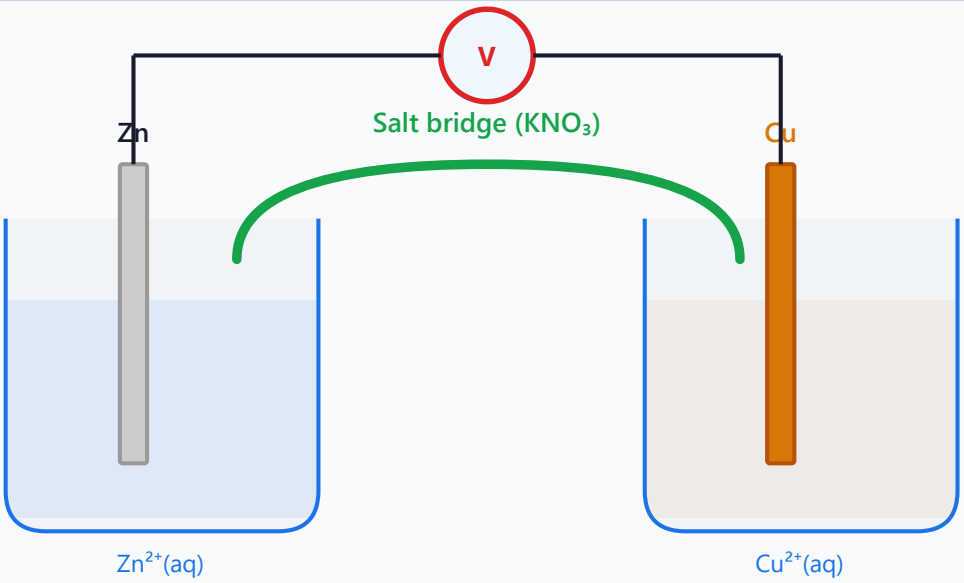
Record: Table of positive/negative results. Use the pattern to identify functional groups.

Error Analysis

Source of Error	Type	Improvement
Dirty glassware	Systematic	Clean thoroughly; Tollens' needs very clean tubes
Insufficient heating	Systematic	Use water bath at correct temperature
Using old Tollens' reagent	Systematic	Prepare fresh immediately before use
False positives from contamination	Systematic	Run blank tests; use fresh reagents

Exam Tip: To distinguish aldehyde from ketone: both give orange ppt with 2,4-DNP, but only aldehydes give a positive Tollens'/Fehling's. Tertiary alcohols do NOT react with $K_2Cr_2O_7$ (no colour change) because they cannot be oxidised without breaking C-C bonds.

⚗ 7: Electrochemical Cells and EMF Measurement



E°(cell) = E°(cathode) - E°(anode) = +0.34 - (-0.76) = +1.10 V

Apparatus

- Beakers, metal electrodes (Zn, Cu, Fe, Mg, etc.)
- 1.0 mol dm^{-3} metal salt solutions
- Salt bridge (KNO_3 soaked filter paper or agar gel)
- High-resistance voltmeter, wires, crocodile clips
- Emery paper

Method

1. Clean electrodes with emery paper to remove oxide.
2. Place each electrode in its corresponding 1.0 mol dm^{-3} metal ion solution.
3. Connect the two half-cells with a salt bridge.
4. Connect electrodes to the high-resistance voltmeter.
5. Record the EMF and note the polarity (which is +, which is -).
6. Test multiple metal combinations and compare with E° values.

Safety

- Metal salt solutions may be toxic — gloves and goggles.

- Dispose of solutions correctly.

Analysis

Calculate: E°(cell) = E°(cathode) - E°(anode). Compare measured with theoretical.

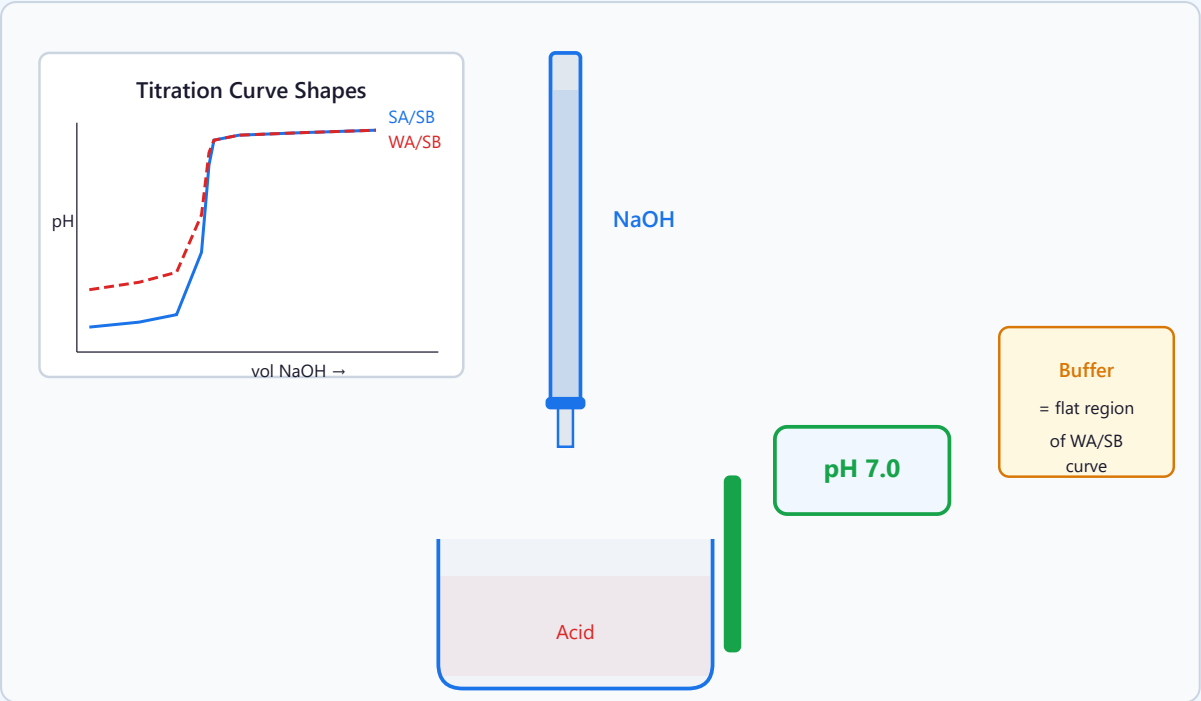
Investigate: Effect of changing concentration on EMF (qualitative Nernst).

Error Analysis

Source of Error	Type	Improvement
Oxide layer on electrodes	Systematic	Clean with emery paper before use
Non-standard concentrations	Systematic	Use volumetric flask to prepare
Salt bridge drying out	Systematic	Fresh salt bridge; good immersion
Voltmeter not high enough resistance	Systematic	Use high-resistance voltmeter or potentiometer

Exam Tip: The more reactive metal is the anode (negative terminal, oxidation). The less reactive is the cathode (positive terminal, reduction). A high-resistance voltmeter is essential because if current flows, the cell does work and the measured EMF is less than the true EMF. Remember: AN OX (anode = oxidation), RED CAT (reduction = cathode).

⚗ 8: pH Curves and Buffer Solutions



Apparatus

- Calibrated pH meter/probe
- Burette with NaOH (or HCl)
- Beaker with acid (or base)
- Magnetic stirrer
- Buffer solutions for calibration

Method

1. Calibrate pH meter with standard buffers.
2. Pipette 25.0 cm^3 of acid into beaker. Record initial pH.
3. Add base from burette in small increments. Record pH after each addition.
4. Use 0.1-0.2 cm^3 increments near the steep equivalence region.
5. Continue well past equivalence until pH plateaus.
6. Plot pH vs. volume to produce the titration curve.
7. **Buffer investigation:** Prepare buffer (HA + NaA). Add small amounts of HCl and NaOH. Show pH barely changes compared to water.

Safety

- NaOH — corrosive, goggles and gloves.
- pH probes — fragile, handle carefully.

Analysis

Identify: Equivalence point, half-equivalence point (pH = pK_a for WA/SB), buffer region.

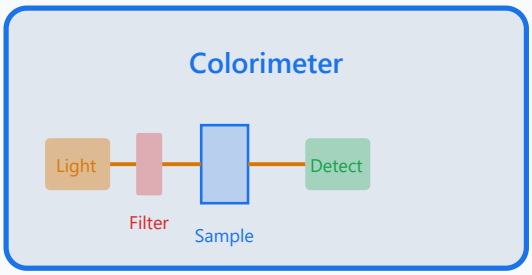
Select indicator: Must have colour change within the steep region of the curve.

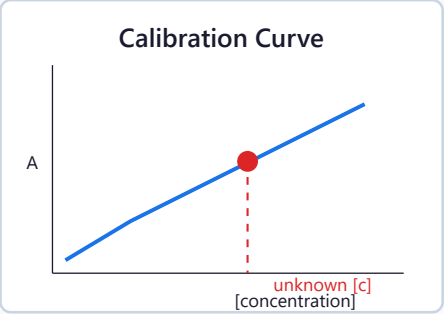
Error Analysis

Source of Error	Type	Improvement
Uncalibrated pH meter	Systematic	Calibrate with 2-3 buffers
Too-large increments near equivalence	Systematic	Use 0.1 cm^3 additions
pH not stabilised	Random	Wait for stable reading; stir continuously
CO_2 absorption	Systematic	Fresh NaOH; keep stoppered

Exam Tip: For weak acid + strong base: initial pH > 1 (partially dissociated), buffer region before equivalence, equivalence pH > 7, half-equivalence pH = pK_a. The buffer region exists because both HA and A^- are present in significant quantities. Use phenolphthalein for this combination.

⚗ 9: Colorimetry and Transition Metal Complexes





A = εcl (Beer-Lambert Law)

Select complementary colour filter for the solution

Blue solution → use orange/red filter | Green solution → use red filter

Apparatus

- Colorimeter with selectable filters, cuvettes
- Standard solutions of known concentration (e.g., $CuSO_4$)
- Volumetric flasks for dilution
- Unknown concentration solution

Method

1. Select the appropriate colour filter (complementary to the solution colour).
2. Zero the colorimeter using distilled water as a blank.
3. Prepare at least 5 standard solutions of known concentration.
4. Measure the absorbance of each standard and the unknown.
5. Plot a calibration curve: absorbance (y) vs. concentration (x).
6. Read the unknown concentration from the graph using its absorbance value.

Safety

- Transition metal salts — wear gloves.
- Handle cuvettes by the top edges only.

Analysis

Plot: Calibration curve (should be linear at low concentrations per Beer-Lambert Law).

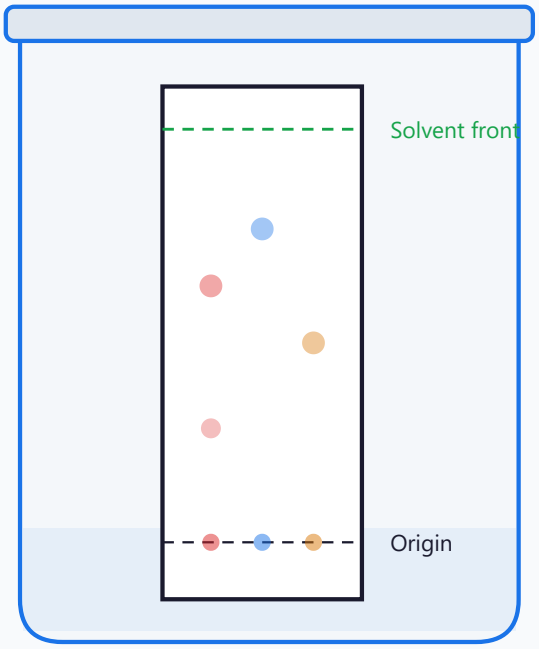
Read: Unknown concentration by interpolation (never extrapolation).

Error Analysis

Source of Error	Type	Improvement
Fingerprints/scratches on cuvette	Systematic	Handle by top; clean with lens tissue; same orientation
Wrong filter	Systematic	Use complementary colour; determine λmax experimentally
Bubbles in solution	Random	Degas solutions; tap to remove bubbles
Standards incorrectly prepared	Systematic	Use volumetric flasks and pipettes

Exam Tip: The calibration curve must pass through the origin (zero concentration = zero absorbance after blanking). Only use values within the range of the calibration standards (interpolation, not extrapolation). If the curve is not linear at high concentrations, the Beer-Lambert Law breaks down — use only the linear portion.

⚗ 10: Chromatography



Rf Value

$$\frac{\text{distance moved by spot}}{\text{distance by solvent front}}$$

Paper chrom.

Partition between mobile (solvent) and stationary (water in paper fibres) phase

Use pencil for origin line (ink dissolves). Spots must be above solvent level.

Apparatus

- TLC plates (silica or alumina) or chromatography paper
- Developing chamber (beaker with watch glass/lid)
- Capillary tubes, pencil, ruler
- Suitable solvent(s)
- UV lamp or ninhydrin/iodine for visualisation

Method

1. Draw a pencil origin line 1 cm from the bottom of the plate/paper.
2. Spot samples (and known references) onto the origin using capillary tubes.
3. Pour solvent into the chamber to 0.5 cm depth (below origin). Allow to saturate.
4. Place the plate/paper in the chamber. Cover. Let solvent rise.
5. Remove when solvent front is near the top. Mark the solvent front immediately.
6. Visualise spots (UV, iodine, or ninhydrin for amino acids).
7. Measure distances and calculate Rf values.

Safety

- Solvents are flammable — no flames, fume cupboard.
- UV lamp — avoid direct exposure to eyes.
- Ninhydrin — irritant, wear gloves.

Analysis

Calculate: Rf = distance by spot / distance by solvent. Compare with reference standards.

TLC: Adsorption chromatography — polar substances have lower Rf (interact more with polar stationary phase).

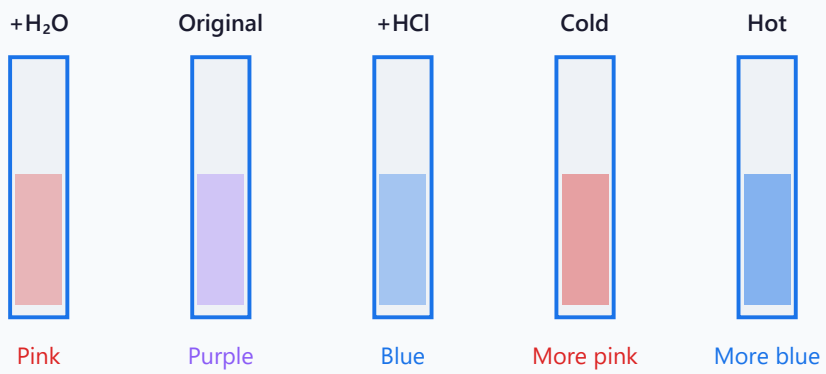
Paper: Partition chromatography — substances partition between water in paper and mobile solvent.

Error Analysis

Source of Error	Type	Improvement
Spots too large / overloaded	Systematic	Small concentrated spots; dry between applications
Origin below solvent level	Systematic	Check depth carefully
Chamber not saturated	Systematic	Line with filter paper; wait before inserting plate
Solvent front not marked in time	Random	Mark immediately on removal

Exam Tip: TLC uses adsorption (interaction with solid stationary phase). Paper chromatography uses partition (between two liquids). In TLC, more polar compounds have lower Rf because they adsorb more strongly to the polar silica. Rf values are only comparable under identical conditions.

⚗ 11: Equilibrium Investigation (Le Chatelier's)



$[Co(H_2O)_6]^{2+}$(pink) + $4Cl^-$ ⇌ $[CoCl_4]^{2-}$(blue) + $6H_2O$ ΔH = +ve

Only temperature changes Kc. Concentration changes shift position, not Kc.

Apparatus

- Test tubes, test tube rack
- $CoCl_2$ solution (equilibrium mixture)
- Concentrated HCl, distilled water
- Water baths (ice and hot water)
- Teat pipettes

Method

1. Prepare cobalt chloride equilibrium mixture (purple at room temperature).
2. Divide into several test tubes.
3. **Add HCl:** Observe shift to blue (increased $[Cl^-]$ shifts equilibrium right).
4. **Add water:** Observe shift to pink (dilutes Cl^-, shifts equilibrium left).
5. **Cool in ice:** Observe shift to pink (exothermic direction favoured).
6. **Heat in hot water:** Observe shift to blue (endothermic direction favoured).

7. Record all observations and explain using Le Chatelier's principle.

Safety

- $CoCl_2$ is toxic and a suspected carcinogen — gloves.
- Conc. HCl is corrosive — fume cupboard.
- Hot water — care with burns.

Analysis

Record: Table of condition changed, direction of shift, colour observed, explanation.

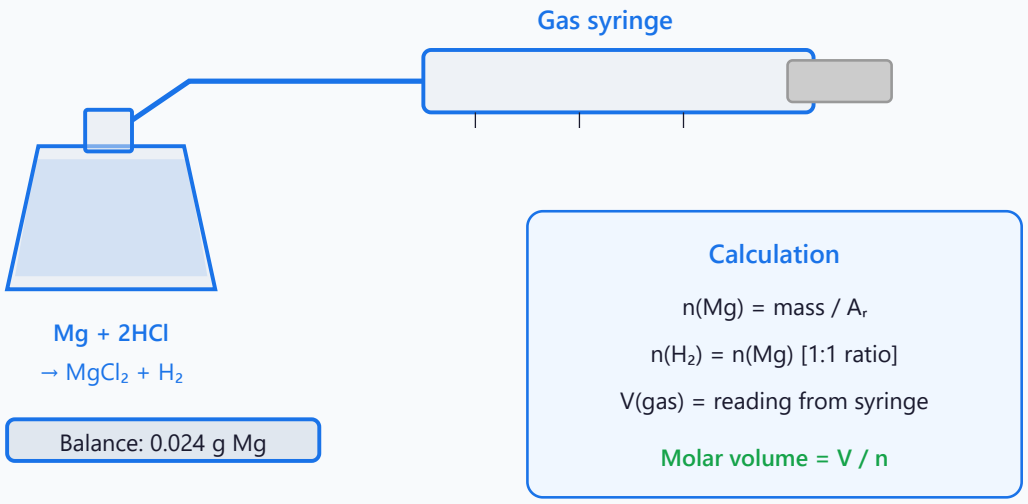
Key: Forward reaction is endothermic. Heating shifts right (endothermic favoured). Adding Cl^- shifts right. Adding H_2O shifts left.

Error Analysis

Source of Error	Type	Improvement
Subjective colour judgement	Random	Use colorimeter for quantitative data
Temperature not maintained	Systematic	Monitor with thermometer; maintain bath
Equilibrium not re-established	Systematic	Allow sufficient time
Uncontrolled amounts added	Systematic	Measure volumes with pipette

Exam Tip: A catalyst does NOT shift the position of equilibrium and does NOT change Kc. It only increases the rate at which equilibrium is reached. In exam questions, always state that Le Chatelier's principle predicts the system opposes the change imposed. Only temperature changes Kc; concentration and pressure changes only shift the position.

⚗ 12: Moles and Gas Volume Determination



Expected: ~24,000 cm^3 mol^{-1} at RTP (room temperature and pressure)

Apparatus

- Conical flask with bung and delivery tube
- Gas syringe (100 cm^3)
- Balance (to 0.001 g)
- Measuring cylinder
- Magnesium ribbon (weighed accurately)
- Excess dilute HCl
- Thermometer and barometer (to record conditions)

Method

1. Weigh a known length of clean Mg ribbon accurately (e.g., ~0.03 g).
2. Place excess dilute HCl (e.g., 20 cm^3 of 1 mol dm^{-3}) in the conical flask.
3. Connect the delivery tube to the gas syringe. Check there are no leaks.
4. Add the Mg ribbon to the flask, quickly insert the bung, and allow the reaction to proceed.
5. Wait until the reaction is complete (Mg fully dissolved, no more gas produced).
6. Record the final volume of H_2 gas in the syringe.

7. Record room temperature and atmospheric pressure.
8. Calculate: n(Mg) = mass/24.3. $n(H_2)$ = n(Mg). Molar gas volume = V(gas)/$n(H_2)$.
9. Alternatively, use pV = nRT to calculate n from p, V, and T, then find M_r.

Safety

- H_2 gas is flammable — no naked flames.
- Dilute HCl is irritant — goggles.
- Gas syringe may pop out — clamp securely.

Analysis

Calculate: Molar gas volume at room conditions. Expected ~24 dm^3 mol^{-1} (24,000 cm^3 mol^{-1}) at ~298 K, 101 kPa.

Alternative: Use pV = nRT to find n, then compare with calculated n from mass.

Error Analysis

Source of Error	Type	Improvement
H_2 gas escaping before sealing	Systematic	Seal quickly; use an inverted flask method
Gas syringe friction	Systematic	Lubricate; tap gently
Mg oxide layer on ribbon	Systematic	Clean with emery paper before weighing
Gas not at room temperature	Systematic	Allow time for gas to cool before reading
H_2 slightly soluble in water/acid	Systematic	Negligible at normal conditions; acknowledged

Exam Tip: The molar gas volume at RTP (room temperature and pressure, ~298 K, 100 kPa) is approximately 24.0 dm^3 mol^{-1}. At STP (273 K, 100 kPa) it is 22.7 dm^3 mol^{-1}. Always state the conditions. Use pV = nRT when exact conditions matter: p in Pa, V in m^3, T in K, R = 8.314.

Worked Example — Water of Crystallisation:

Mass of crucible = 22.50 g. Mass of crucible + hydrated salt = 27.48 g. Mass after heating to constant mass = 25.28 g.

Mass of water lost = 27.48 − 25.28 = 2.20 g → $n(H_2O)$ = 2.20 / 18.0 = 0.122 mol
Mass of anhydrous salt = 25.28 − 22.50 = 2.78 g → $n(MgSO_4)$ = 2.78 / 120.4 = 0.0231 mol
Ratio = 0.122 / 0.0231 = 5.28 ≈ **5**, so formula is **$MgSO_4 \cdot 5H_2O$** (but the accepted value is 7 — incomplete dehydration or experimental error may account for this discrepancy).